XMAKE SURFACE FINISHING

真空镀（PVD）
Vacuum Plating (PVD)

中英文对照审核稿 / Bilingual Review Draft

Primary keyword

pvd coating services

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中文审核稿

Meta Title：PVD Coating Services | 真空镀与功能薄膜 - Xmake

Meta Description：Xmake 提供面向金属、选定塑料、玻璃和陶瓷零件的 PVD coating services，覆盖装饰真空金属化、磁控溅射、离子镀及功能性硬质薄膜。提交材料、图纸、颜色、遮蔽和性能要求，获取工艺评估与报价。

Primary Keyword：pvd coating services

Secondary Keywords：vacuum plating services, PVD coating, vacuum metallizing, decorative PVD coating, PVD coating for plastic, metal PVD coating, magnetron sputtering, ion plating

Search Intent：交易型 / 商业调研型。买家正在比较真空镀与电镀、喷涂等方案，确认基材兼容性、外观、耐磨性、尺寸影响、遮蔽和量产条件，并准备提交 RFQ。

H1：PVD 真空镀服务

Xmake 为需要金属质感、指定颜色、反射效果或功能性薄膜的零件提供 pvd coating services。项目可从零件制造延伸到表面准备、遮蔽、装夹、真空沉积、保护面层和检验，由 Xmake 的自营工厂、自有设备及制造资源协同完成。

“真空镀”是市场上常用的宽泛名称。本页重点介绍物理气相沉积（PVD）：在真空环境中，通过蒸发、溅射或离子化方式将固体材料转化为气相，并在零件表面凝结形成薄膜。具体路线取决于基材、几何、目标外观、使用环境和性能要求。

PVD 是薄膜工艺，不会自动填平深划痕、刀纹、缩痕、焊接缺陷或明显粗糙度。最终效果从基材表面开始，因此报价时应同时确定前处理、样件标准、可见面、遮蔽面和验收方法。

- 装饰效果：银色、金色、玫瑰金、枪色、黑色、青铜色及项目指定颜色。
- 功能目标：耐磨、表面硬度、低摩擦、反射或导电等项目定义性能。
- 供应链整合：零件制造、抛光、清洗、PVD、保护涂层、检验和交付可按同一项目计划管理。

H2：PVD 真空镀规格

规格	说明
主要用途	装饰金属外观、颜色控制、反射层、耐磨或其他项目定义的功能薄膜
典型工艺	真空蒸发镀、磁控溅射、阴极电弧 / 离子镀、反应性 PVD
适用基材	不锈钢、工具钢、铝合金、铜合金、选定塑料、玻璃和陶瓷；按具体工艺评估
外观选择	高亮、缎面、哑光、拉丝底纹、镜面底纹及多种颜色；最终效果受基材和前处理影响
尺寸影响	PVD 为薄膜沉积，但关键配合、螺纹、密封面、孔位和电接触区仍需图纸定义与遮蔽评审
覆盖特征	以视线可达沉积为主；深孔、内腔、背面、阴影区和复杂凹槽需要装夹与覆盖评估
质量确认	可按项目使用外观样板、颜色 / 光泽、膜厚、附着力、硬度、耐磨或其他约定测试

H2：什么是 PVD 真空镀?

PVD 是在低压真空环境中形成薄膜的一组工艺。镀层材料从固体源释放，经真空空间传输后在基材表面凝结。常见能量来源和沉积机制包括电阻或电子束蒸发、磁控溅射以及阴极电弧。加入氮、氧等反应气体时，还可形成项目指定的化合物薄膜。

PVD 与传统湿式电镀的核心区别在于沉积环境和材料传输方式。PVD
不依赖把零件浸入金属盐电镀液完成沉积，但前处理、清洁、装夹、真空度、温度和工艺稳定性同样决定结果。

对装饰件而言，PVD 可在合适的基材和底层上形成连续的金属质感或颜色。对功能件而言，可选用硬质或低摩擦薄膜改善表面性能。两类目标不能只用颜色名称代替技术要求：装饰件应定义色样、光泽、可见面和使用环境；功能件应定义膜层体系、厚度、硬度、附着力、摩擦、耐磨或其他验收指标。

H2：Xmake 提供的 PVD 工艺路线

H3：装饰真空金属化

装饰真空金属化常用于需要亮银、金属镜面或半反射效果的塑料与金属零件。塑料件通常需要底涂、表面活化或等离子处理，以改善表面平整度和附着力；沉积后可根据耐刮、颜色和环境要求增加透明或着色保护面层。

适用方向：消费产品外壳、装饰盖、旋钮、标牌、照明反射部件和展示件。
控制重点：塑料牌号、脱模剂残留、底涂兼容性、热负荷、挂点、边缘覆盖和面层耐久性。

H3：磁控溅射

磁控溅射利用等离子体轰击靶材，使靶材原子或粒子释放并沉积到零件表面。该路线适合需要较稳定薄膜组成、装饰颜色、金属导电层或光学 / 反射功能的项目，也可用于金属、选定塑料、玻璃和陶瓷。

适用方向：精密外观件、电子部件、光学与反射部件、薄金属层和多层薄膜。
控制重点：靶材、反应气体、基材温度、膜层均匀性、装夹旋转、遮蔽和批次色差。

H3：阴极电弧与离子镀

阴极电弧和离子镀适合在金属零件上形成结合力较强的装饰或功能薄膜。可按项目开发金色、黑色、枪色等装饰外观，也可形成以耐磨、硬度或低摩擦为重点的硬质薄膜。

适用方向：五金件、工具部件、模具零件、耐磨组件和高接触频率装饰件。
控制重点：基材硬度、回火稳定性、沉积温度、表面粗糙度、刃口状态、膜层体系和使用载荷。

H3：反应性与多层 PVD

反应性 PVD 在沉积过程中引入工艺气体，形成氮化物、氧化物或其他项目指定薄膜。多层或梯度结构可用于平衡附着力、颜色、硬度、内应力和表面功能。

适用方向：指定颜色、硬质保护、耐磨、低摩擦和多功能表面。
控制重点：基材兼容性、膜层结构、总厚度、界面结合、热影响和测试标准。

H2：PVD 颜色与表面效果

目标效果	常见外观	适用说明	需要确认
银色 / 铬色	高亮、镜面、缎面或哑光金属感	装饰金属化、溅射或项目指定薄膜	底材纹理、亮度、保护面层和色样
金色 / 香槟金	暖金、浅金或香槟色	装饰五金、消费产品和展示件	标准色样、光泽、批次色差和使用环境
玫瑰金 / 铜色	暖红金属色	装饰件和品牌外观件	色相范围、底层颜色和耐变色要求

目标效果	常见外观	适用说明	需要确认
黑色 / 枪色	深灰、黑色或烟灰金属感	装饰件、五金件及功能薄膜	光泽、指纹要求、摩擦区和边缘状态
青铜 / 古铜	暖棕或深铜色	设计五金、家居和展示件	色样、纹理、做旧效果和面层
半透明 / 半反射	部分透光或半镜面效果	灯具、显示窗口和装饰组件	透过率、反射率、基材与光源条件

颜色名称不是统一的技术标准。同一“金色”会因膜层体系、基材纹理、表面粗糙度、观察角度、光源和保护面层而变化。量产项目应使用实物色板、限度样或经批准的首件作为验收依据。

H2：适用材料

金属

不锈钢、工具钢、模具钢、碳钢、铝合金、铜及铜合金均可根据膜层和温度路线进行评估。金属件应在沉积前完成去油、清洗、抛光或纹理处理；氧化层、锈蚀、焊渣和残余抛光膏会直接影响附着与外观。

塑料

ABS、PC/ABS、PC、PA 及其他工程塑料可按牌号、填料、耐温、放气和表面状态评估。塑料通常需要底涂、等离子活化或其他前处理。不同塑料并非自动共用同一工艺窗口，量产前应完成样件验证。

玻璃与陶瓷

玻璃和陶瓷可用于装饰、反射、导电或其他薄膜项目，但需要确认清洁度、热膨胀、附着、装夹和边缘风险。透明基材还应定义观察方向、透过率、反射率和颜色要求。

H2：基材、工艺与结果匹配

基材 / 零件条件	建议路线	可实现的结果	关键控制点
抛光不锈钢外观件	装饰溅射或离子镀	高亮或彩色金属外观	抛光等级、清洗、挂点、色样和防指纹要求
工具钢或模具零件	硬质 PVD	耐磨、硬度或低摩擦表面	基材硬度、回火温度、刃口、膜层体系和载荷
ABS / PC-ABS 外壳	底涂 + 真空金属化 + 保护面层	铬色、银色或着色金属感	塑料牌号、底涂、放气、热负荷、附着和面层
铝合金零件	溅射、离子镀或项目指定路线	装饰颜色或功能薄膜	合金、原始表面、氧化层、温度和尺寸要求
玻璃 / 陶瓷组件	溅射或项目指定 PVD	反射、颜色、导电或功能薄膜	清洁、附着、透光 / 反射要求和装夹

H2：PVD 的优势与设计限制

优势

- 外观范围广：可获得从镜面金属感到哑光、拉丝和彩色薄膜的多种效果。
- 薄膜尺寸影响小：适合需要保留几何精度的零件，但关键配合仍需图纸控制。
- 可兼顾装饰与功能：不同膜层可服务于颜色、反射、耐磨、硬度、低摩擦或导电目标。
- 可覆盖多类基材：金属、选定塑料、玻璃和陶瓷可采用不同前处理与沉积路线。

- 工艺整合：Xmake 可将零件制造、表面准备、PVD、后处理和检验纳入同一交付流程。

设计与工艺限制

- PVD 主要是视线可达工艺。深孔、狭窄内腔、背面和遮挡区的沉积量可能明显降低。
- 薄膜会复制原始表面。深划痕、孔隙、缩痕、刀纹和焊接缺陷不会被自动遮盖。
- 尖锐边缘、薄壁和复杂装夹会影响局部温度、膜厚、挂点和外观一致性。
- 螺纹、压配合、轴承位、密封面和电接触面应在图纸中标出是否遮蔽。
- 塑料件需要评估耐温、放气、脱模剂、涂层附着和底涂体系。
- 功能膜的性能取决于完整组合：基材、前处理、膜层体系、厚度、载荷和环境缺一不可。

H2：PVD 与其他表面处理如何选择

工艺	主要特点	更适合的项目	采购时的判断重点
PVD 真空镀	真空薄膜；装饰与功能路线丰富	金属质感、指定颜色、反射、耐磨或低摩擦需求	基材、视线覆盖、温度、膜层体系和色样
电镀	湿式电化学沉积，可形成较厚金属层	防腐、导电、装饰或尺寸修复等指定体系	基材、前处理、厚度、氢脆和环保要求
喷涂 / 喷粉	有机涂层，颜色与纹理选择广	较厚保护层、颜色覆盖和大面积外观	附着、膜厚、固化温度、边缘和遮蔽
阳极氧化	铝及部分金属的转化 / 氧化层	铝件耐蚀、颜色和表面功能	合金、颜色、尺寸变化和导电区域

选择不应只依据外观照片。应从基材、零件几何、目标寿命、接触载荷、环境、尺寸公差、成本和批量共同判断。

H2：Xmake PVD 真空镀流程

1. 需求与图纸评审：确认材料、数量、目标颜色、膜层功能、可见面、遮蔽面、配合和测试要求。
2. 工艺与样件规划：选择蒸发、溅射、离子镀或多层路线，并定义基材准备、样板和验收边界。
3. 零件制造与表面准备：完成机加工、成型、去毛刺、抛光、纹理、清洗和必要的底涂或活化。
4. 遮蔽与装夹：保护螺纹、密封面、配合面和电接触区，并规划挂点、旋转和视线覆盖。
5. 真空沉积：按膜层体系控制真空、功率、气体、温度、时间和零件运动。
6. 后处理：按需要增加保护面层、固化、清洁或其他项目指定处理。
7. 检验与交付：依据批准样、图纸和检验计划检查外观、颜色、覆盖、关键尺寸及项目定义性能。

H2：为什么选择 Xmake 的 PVD Coating Services

自营工厂与自有设备基础

Xmake 以自营工厂、自有设备和制造资源为项目执行基础，可将零件制造和表面处理要求放在同一工程与质量框架内管理。复杂项目可从图纸、材料、加工路线开始评审，而不是等零件完成后再单独处理外观问题。

从基材到膜层的工程协同

PVD 的结果由基材、表面准备、遮蔽、装夹和膜层共同决定。Xmake
可在制造阶段识别可见面、挂点、边缘、深孔、螺纹和配合风险，降低返工与批次偏差。

装饰与功能目标分开定义

装饰项目以色样、光泽、纹理和外观边界为核心；功能项目以膜层体系、厚度、附着、硬度、耐磨或摩擦指标为核心。分开定义可避免用“PVD”一个名称代替实际验收标准。

面向量产的样件与检验路径

首件、限度样、遮蔽图和检验计划可以在批量前锁定。若项目涉及颜色、反射或透明效果，应在实际基材和最终表面准备条件下批准样件。

H2：常见问题

PVD 真空镀和电镀有什么区别?

PVD 在真空中把固体源材料转化并沉积成薄膜；电镀通常在电解液中通过电化学反应沉积金属。两者在适用基材、膜厚、覆盖方式、颜色、性能和前处理方面不同，应按零件目标选择。

塑料可以做 PVD 吗?

可以。选定塑料可通过清洗、底涂、等离子活化和真空金属化形成金属外观。需要确认树脂牌号、填料、耐温、放气、脱模剂残留、附着和保护面层。

PVD 可以做哪些颜色?

常见方向包括银色、金色、香槟金、玫瑰金、铜色、青铜色、枪色和黑色。实际颜色由膜层体系、基材纹理、光泽和面层决定，量产应使用批准色样。

PVD 会影响零件尺寸吗?

PVD 是薄膜沉积，尺寸影响通常小于较厚的喷涂或部分电镀体系，但不能视为零影响。精密配合、螺纹、密封面、孔径和轴承位仍需定义膜厚与遮蔽。

PVD 能覆盖深孔和内腔吗?

覆盖受视线、零件旋转、装夹和几何影响。深孔、狭窄内腔、背面和遮挡区域的膜厚可能低于外表面，需要在图纸评审时确认。

PVD 表面是否耐刮耐磨?

取决于基材、膜层类型、厚度、附着、表面硬度、载荷和环境。装饰金属化与硬质功能 PVD 不是同一性能等级，应按用途定义测试。

报价需要提供什么?

请提供 2D / 3D 图纸、材料与牌号、数量、目标颜色或实物色样、可见面、遮蔽区域、关键公差、膜层功能、使用环境和测试要求。塑料件还应说明树脂、填料和现有表面状态。

CTA：获取 PVD 真空镀报价

发送图纸、材料、数量、目标颜色或功能、关键尺寸、遮蔽区域和验收要求。Xmake 将评估基材准备、PVD 路线、装夹覆盖、样件方案和批量交付条件。

提交您的 PVD Coating RFQ

Suggested Internal Links

- Surface Finishing Services
- CNC Machining Services
- Injection Molding Services

- Electrophoretic Coating Services
- Tumbling and Deburring Services
- CNC Engraving Services
- Water Transfer Printing Services

Human Verification Notes

- 确认实际基材牌号、热处理状态、原始粗糙度和现有表面层。
- 确认装饰目标采用实物色样、颜色编号还是批准首件作为验收标准。
- 确认膜层路线、材料体系、总厚度和允许的批次色差。
- 确认塑料件底涂、等离子处理、耐温、放气和附着验证方案。
- 确认深孔、内腔、背面和阴影区的覆盖边界，以及零件旋转与装夹方案。
- 确认螺纹、密封面、压配合、电接触区、挂点和非镀区域的遮蔽图。
- 确认功能件所需的附着力、硬度、耐磨、摩擦、耐腐蚀或其他测试方法及接受值。
- 确认展示图片均标注真实材料、前处理、膜层、颜色和保护面层，避免仅凭图片推断固定效果。

Technical Reference Basis

- Xmake 官方网站：自营工厂、制造能力与表面处理服务基础。https://www.xmake.com/
- Xmake Our Factory：自营工厂与设备信息。https://xmake.com/about/our-factory/
- Xmake Surface Finishing：表面处理服务范围。https://xmake.com/surface-finishing/
- Ionbond, Physical Vapor Deposition：PVD 定义、溅射 / 阴极电弧、清洗、视线覆盖与基材要求。https://www.ionbond.com/technology/pvd/
- Marmillon, Vacuum Metallization：塑料真空金属化、蒸发、磁控溅射与等离子前处理。https://marmillon.com/en/vacuum-metallization/
- Acree Technologies, PVD Coatings for Plastics：塑料等离子前处理与多层膜结构。https://acreetech.com/plastics/

English Review Draft

Meta Title: PVD Coating Services | Vacuum Plating & Functional Films - Xmake

Meta Description: Xmake provides PVD coating services for metal, selected plastic, glass, and ceramic parts, including decorative vacuum metallizing, magnetron sputtering, ion plating, and functional hard films. Submit material, drawings, color, masking, and performance requirements for process review and quotation.

Primary Keyword: pvd coating services

Secondary Keywords: vacuum plating services, PVD coating, vacuum metallizing, decorative PVD coating, PVD coating for plastic, metal PVD coating, magnetron sputtering, ion plating

Search Intent: Transactional / commercial investigation. The buyer is comparing vacuum plating with electroplating, painting, or other finishes and needs to confirm substrate compatibility, appearance, wear performance, dimensional impact, masking, and production conditions before submitting an RFQ.

H1: PVD Coating Services

Xmake provides pvd coating services for parts requiring a metallic appearance, controlled color, reflective behavior, or a functional thin film. A project can extend from part manufacturing through surface preparation, masking, fixturing, vacuum deposition, protective topcoating, inspection, and delivery, supported by Xmake’ s self-operated factories, owned equipment, and manufacturing resources.

“Vacuum plating” is a broad commercial term. This page focuses on physical vapor deposition (PVD), in which a solid coating material is evaporated, sputtered, or ionized in a vacuum and condensed as a thin film on the part surface. The correct route depends on the substrate, geometry, target appearance, service environment, and required performance.

PVD is a thin-film process. It does not automatically hide deep scratches, machining marks, sink marks, weld defects, or pronounced roughness. The final result begins with the substrate, so pretreatment, approved samples, visible surfaces, masked areas, and acceptance methods should be defined at the quotation stage.

- Decorative outcomes: silver, gold, rose gold, gunmetal, black, bronze, and project-defined colors.
- Functional objectives: wear resistance, surface hardness, low friction, reflection, conductivity, or other project-defined properties.
- Supply-chain integration: part production, polishing, cleaning, PVD, protective coating, inspection, and delivery can be managed under one project plan.

H2: PVD Coating Specifications

Specification	Details
Primary uses	Decorative metallic appearance, color control, reflective films, wear resistance, or other project-defined functional films
Typical processes	Vacuum evaporation, magnetron sputtering, cathodic arc / ion plating, and reactive PVD
Suitable substrates	Stainless steel, tool steel, aluminum alloys, copper alloys, selected plastics, glass, and ceramics, reviewed by process
Appearance options	Bright, satin, matte, brushed-base, mirror-base, and multiple colors; final appearance depends on the substrate and preparation

Specification	Details
Dimensional impact	PVD forms a thin film, but critical fits, threads, sealing faces, holes, and electrical contact areas still require drawing definition and masking review
Coverage behavior	Primarily line-of-sight deposition; deep holes, internal cavities, back sides, shadowed zones, and complex recesses require fixturing and coverage review
Quality confirmation	Appearance masters, color / gloss, thickness, adhesion, hardness, wear, or other agreed tests can be applied by project

H2: What Is PVD Coating?

PVD is a family of thin-film processes performed in a low-pressure vacuum. Coating material is released from a solid source, transported through the vacuum, and condensed on the substrate. Common energy sources and deposition mechanisms include resistance or electron-beam evaporation, magnetron sputtering, and cathodic arc. When process gases such as nitrogen or oxygen are introduced, project-defined compound films can also be formed.

The principal difference between PVD and conventional wet electroplating is the deposition environment and material-transfer mechanism. PVD does not rely on immersing the part in a metal-salt plating bath, but pretreatment, cleanliness, fixturing, vacuum conditions, temperature, and process stability are still decisive.

For decorative parts, PVD can create a continuous metallic appearance or color on a suitable substrate and base layer. For functional parts, hard or low-friction films can improve surface behavior. These goals need different acceptance criteria. Decorative work should define the color master, gloss, visible surfaces, and service environment. Functional work should define the coating system, thickness, hardness, adhesion, friction, wear, or other test metrics.

H2: PVD Processes We Offer

H3: Decorative Vacuum Metallizing

Decorative vacuum metallizing is used for plastic and metal parts requiring bright silver, mirror-like metallic, or semi-reflective effects. Plastic parts commonly need a basecoat, surface activation, or plasma treatment to improve smoothness and adhesion. A clear or tinted protective topcoat can be added after deposition when scratch resistance, color, or environmental durability requires it.

Typical uses: consumer housings, decorative covers, knobs, nameplates, lighting reflectors, and display components. Key controls: plastic grade, mold-release residue, basecoat compatibility, thermal load, rack points, edge coverage, and topcoat durability.

H3: Magnetron Sputtering

Magnetron sputtering uses plasma to bombard a target, releasing atoms or particles that deposit on the part. It is suitable for projects requiring controlled film composition, decorative color, a metallic conductive layer, or optical / reflective behavior, and can be used on metals, selected plastics, glass, and ceramics.

Typical uses: precision appearance parts, electronic components, optical and reflective components, thin metallic layers, and multilayer films. Key controls: target material, reactive gas, substrate temperature, film uniformity, part rotation, masking, and batch color variation.

H3: Cathodic Arc and Ion Plating

Cathodic arc and ion plating are used to form strongly bonded decorative or functional films on metal parts. Projects may target gold, black, gunmetal, and other decorative appearances, or hard films focused on wear, hardness, or low friction.

Typical uses: hardware, tool components, mold parts, wear components, and frequently handled decorative parts. Key controls: substrate hardness, tempering stability, deposition temperature, surface roughness, edge condition, coating architecture, and operating load.

H3: Reactive and Multilayer PVD

Reactive PVD introduces process gases during deposition to form nitrides, oxides, or other project-defined films. Multilayer or graded structures can balance adhesion, color, hardness, internal stress, and surface function.

Typical uses: specified colors, hard protection, wear resistance, low friction, and multifunctional surfaces. Key controls: substrate compatibility, coating architecture, total thickness, interface bonding, thermal influence, and test standard.

H2: PVD Colors and Surface Effects

Target effect	Common appearance	Application guidance	Confirm before production
Silver / chrome-like	Bright, mirror, satin, or matte metallic	Decorative metallizing, sputtering, or project-defined film	Base texture, brightness, protective topcoat, and approved master
Gold / champagne	Warm gold, light gold, or champagne	Decorative hardware, consumer products, and display parts	Color master, gloss, batch variation, and service environment
Rose gold / copper	Warm red metallic tone	Decorative and branded appearance parts	Hue window, base color, and tarnish-resistance requirement
Black / gunmetal	Dark gray, black, or smoky metallic	Decorative hardware and functional films	Gloss, fingerprint requirement, contact zones, and edge condition
Bronze / antique bronze	Warm brown or deep copper tone	Design hardware, home products, and display pieces	Color master, texture, antiquing effect, and topcoat
Translucent / semi-reflective	Partial transmission or one-way-mirror effect	Lighting, display windows, and decorative assemblies	Transmission, reflection, substrate, and light-source conditions

Color names are not universal technical standards. The same “gold” changes with coating architecture, substrate texture, roughness, viewing angle, light source, and protective topcoat. Production work should use a physical color master, limit samples, or an approved first article for acceptance.

H2: Compatible Materials

Metals

Stainless steel, tool and mold steels, carbon steel, aluminum alloys, copper, and copper alloys can be reviewed against the selected film and temperature route. Metal parts should be degreased, cleaned, polished, or textured before deposition. Oxides, corrosion, weld residue, and polishing compound directly affect adhesion and appearance.

Plastics

ABS, PC/ABS, PC, PA, and other engineering plastics can be reviewed by exact grade, filler, heat resistance, outgassing, and surface condition. Plastics commonly require a basecoat, plasma activation, or other pretreatment. Different resins do not automatically share one process window, so sample validation should precede production.

Glass and Ceramics

Glass and ceramics can support decorative, reflective, conductive, or other thin-film applications. Cleanliness, thermal expansion, adhesion, fixturing, and edge risk require review. Transparent substrates should also define viewing direction, transmission, reflection, and color requirements.

H2: Matching Substrate, Process, and Result

Substrate / part condition	Recommended route	Achievable result	Key controls
Polished stainless appearance part	Decorative sputtering or ion plating	Bright or colored metallic appearance	Polish level, cleaning, rack points, color master, and fingerprint requirement
Tool steel or mold component	Hard PVD	Wear-resistant, hard, or low-friction surface	Substrate hardness, tempering temperature, edges, coating system, and load
ABS / PC-ABS housing	Basecoat + vacuum metallizing + protective topcoat	Chrome-like, silver, or tinted metallic appearance	Resin grade, basecoat, outgassing, thermal load, adhesion, and topcoat
Aluminum-alloy part	Sputtering, ion plating, or project-defined route	Decorative color or functional film	Alloy, original surface, oxide layer, temperature, and dimensions
Glass / ceramic component	Sputtering or project-defined PVD	Reflective, colored, conductive, or functional film	Cleaning, adhesion, transmission / reflection requirement, and fixturing

H2: Advantages and Design Limits of PVD

Advantages

- Broad appearance range: from mirror-like metallic effects to matte, brushed-base, and colored films.
- Low dimensional impact: useful for parts that must retain geometry, while critical fits still need drawing control.
- Decorative and functional options: films can support color, reflection, wear, hardness, low friction, or conductivity objectives.
- Multiple substrate classes: metals, selected plastics, glass, and ceramics can use different pretreatment and deposition routes.
- Integrated execution: Xmake can place manufacturing, preparation, PVD, post-treatment, and inspection in one delivery workflow.

Design and Process Limits

- PVD is primarily a line-of-sight process. Deposition can be reduced in deep holes, narrow cavities, back sides, and shadowed areas.
- A thin film reproduces the starting surface. Deep scratches, porosity, sink marks, machining marks, and weld defects are not automatically hidden.
- Sharp edges, thin walls, and complex fixtures affect local temperature, film distribution, rack marks, and appearance consistency.
- Threads, press fits, bearing seats, sealing faces, and electrical contacts should be identified for masking on the drawing.

- Plastic parts require review of temperature resistance, outgassing, mold release, coating adhesion, and basecoat compatibility.
- Functional performance comes from the complete system: substrate, pretreatment, film architecture, thickness, load, and environment.

H2: PVD Compared with Other Finishes

Process	Main characteristics	Better suited to	Purchase decision factors
PVD vacuum coating	Vacuum thin film with decorative and functional routes	Metallic appearance, controlled color, reflection, wear, or low friction	Substrate, line-of-sight coverage, temperature, film system, and color master
Electroplating	Wet electrochemical deposition; can form a thicker metallic layer	Specified corrosion, conductivity, decoration, or dimensional build	Substrate, pretreatment, thickness, hydrogen embrittlement, and environmental requirements
Painting / powder coating	Organic coating with wide color and texture selection	Thicker protection, color coverage, and large visible areas	Adhesion, film thickness, cure temperature, edges, and masking
Anodizing	Conversion / oxide layer for aluminum and selected metals	Aluminum corrosion resistance, color, and surface function	Alloy, color, dimensional change, and conductive zones

Selection should not be based on appearance photos alone. Substrate, geometry, required life, contact load, environment, tolerance, cost, and volume should be considered together.

H2: Xmake PVD Coating Process

1. Requirement and drawing review: Confirm material, quantity, target color, film function, visible faces, masking, fits, and test requirements.
2. Process and sample planning: Select evaporation, sputtering, ion plating, or a multilayer route; define preparation, sample standards, and acceptance limits.
3. Part manufacturing and preparation: Complete machining or molding, deburring, polishing, texturing, cleaning, and any required basecoat or activation.
4. Masking and fixturing: Protect threads, sealing faces, fits, and electrical contacts; plan rack points, rotation, and line-of-sight coverage.
5. Vacuum deposition: Control vacuum, power, gas, temperature, time, and part movement for the selected film architecture.
6. Post-treatment: Add protective topcoating, curing, cleaning, or other project-defined treatment when required.
7. Inspection and delivery: Check appearance, color, coverage, critical dimensions, and project-defined performance against the approved sample, drawing, and inspection plan.

H2: Why Choose Xmake for PVD Coating Services

Self-operated Factories and Owned Equipment

Xmake uses self-operated factories, owned equipment, and manufacturing resources as the execution foundation. Part production and finishing requirements can be managed within one engineering and quality framework. Complex work can begin with drawing, material, and manufacturing-route review instead of treating appearance as a separate issue after parts are complete.

Engineering Coordination from Substrate to Film

PVD results are controlled by substrate condition, surface preparation, masking, fixturing, and coating architecture. Xmake can identify visible faces, rack points, edges, deep holes, threads, and fit risks during manufacturing, reducing rework and batch variation.

Separate Definition of Decorative and Functional Goals

Decorative projects are controlled by color masters, gloss, texture, and appearance boundaries. Functional projects are controlled by coating architecture, thickness, adhesion, hardness, wear, or friction metrics. Separating these goals prevents the single term “PVD” from replacing real acceptance criteria.

Production-oriented Sampling and Inspection

First articles, limit samples, masking drawings, and inspection plans can be established before volume production. Color, reflective, or translucent work should be approved on the actual substrate with the final surface-preparation condition.

H2: Frequently Asked Questions

What is the difference between PVD and electroplating?

PVD converts material from a solid source and deposits it as a film in a vacuum. Electroplating normally deposits metal electrochemically in an electrolyte. They differ in suitable substrates, thickness, coverage behavior, color, performance, and pretreatment, so selection should follow the part requirements.

Can plastic parts receive PVD coating?

Yes. Selected plastics can receive a metallic appearance through cleaning, basecoating, plasma activation, and vacuum metallizing. Resin grade, filler, heat resistance, outgassing, mold-release residue, adhesion, and protective topcoat must be considered.

What colors are available for PVD?

Common directions include silver, gold, champagne, rose gold, copper, bronze, gunmetal, and black. Actual color depends on coating architecture, substrate texture, gloss, and topcoat. Production should use an approved physical master.

Does PVD change part dimensions?

PVD is a thin-film process and normally has less dimensional impact than thicker paint or some plating systems, but it is not zero. Precision fits, threads, sealing faces, hole diameters, and bearing seats still require thickness and masking definition.

Can PVD coat deep holes and internal cavities?

Coverage depends on line of sight, part rotation, fixturing, and geometry. Film thickness in deep holes, narrow cavities, back sides, and shadowed areas can be lower than on exposed surfaces and should be reviewed during drawing assessment.

Is a PVD surface scratch- and wear-resistant?

That depends on the substrate, coating type, thickness, adhesion, surface hardness, load, and environment. Decorative vacuum metallizing and hard functional PVD do not have the same performance level; testing must follow the intended use.

What information is needed for a quotation?

Provide 2D / 3D drawings, material and grade, quantity, target color or physical master, visible faces, masked areas, critical tolerances, film function, service environment, and test requirements. For plastics, include resin, filler, and current surface condition.

CTA: Request a PVD Coating Quote

Send your drawings, material, quantity, target color or function, critical dimensions, masked areas, and acceptance requirements. Xmake will review substrate preparation, the PVD route, fixturing and coverage, sample planning, and production conditions.

Submit Your PVD Coating RFQ

Suggested Internal Links

- Surface Finishing Services
- CNC Machining Services
- Injection Molding Services
- Electrophoretic Coating Services
- Tumbling and Deburring Services
- CNC Engraving Services
- Water Transfer Printing Services

Human Verification Notes

- Confirm the exact substrate grade, heat-treatment condition, starting roughness, and existing surface layer.
- Confirm whether decorative acceptance uses a physical master, color code, or approved first article.
- Confirm the deposition route, film material system, total thickness, and permitted batch color variation.
- Confirm the basecoat, plasma treatment, heat resistance, outgassing, and adhesion-validation route for plastic parts.
- Confirm coverage limits in deep holes, cavities, back sides, and shadow zones, plus part rotation and fixturing.
- Confirm masking drawings for threads, sealing faces, press fits, electrical contacts, rack points, and no-coat zones.
- Confirm required adhesion, hardness, wear, friction, corrosion, or other test methods and acceptance values for functional parts.
- Confirm that production images identify the actual material, preparation, film, color, and protective topcoat rather than implying a fixed result from appearance alone.

Technical Reference Basis

- Xmake official website: self-operated factories, manufacturing capability, and surface-finishing foundation. https://www.xmake.com/
- Xmake Our Factory: self-operated factory and equipment information. https://xmake.com/about/our-factory/
- Xmake Surface Finishing: surface-finishing service range. https://xmake.com/surface-finishing/
- Ionbond, Physical Vapor Deposition: PVD definition, sputtering / cathodic arc, cleaning, line-of-sight coverage, and substrate requirements. https://www.ionbond.com/technology/pvd/
- Marmillon, Vacuum Metallization: plastic vacuum metallizing, evaporation, magnetron sputtering, and plasma pretreatment. https://marmillon.com/en/vacuum-metallization/
- Acree Technologies, PVD Coatings for Plastics: plasma pretreatment for plastics and multilayer film architecture. https://acreetech.com/plastics/